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Circulating Tumor DNA: A Pan-Cancer Biomarker in Solid Tumors with Prognostic and Predictive Value

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Keywords

ctDNA, cancer, biomarker

Abstract

Circulating tumor DNA (ctDNA), often referred to as a liquid biopsy, represents a promising biomarker in the management of both localized and advanced solid tumors. It has garnered significant attention due to its potential to inform prognosis and guide therapeutic decisions. The clinical utility of ctDNA spans early cancer detection, minimal residual disease identification, recurrence surveillance, treatment monitoring, and precision oncology treatment decision-making in the advanced setting. Unlike conventional radiological assessments, the short half-life of ctDNA allows for more timely insights into disease dynamics. Several technological approaches are available to measure ctDNA, including next-generation sequencing and droplet digital polymerase chain reaction, although their clinical accuracy depends on multiple biological and technical factors. This review evaluates current evidence surrounding ctDNA's utility in early and advanced solid tumors.

INTRODUCTION

Circulating tumor DNA (ctDNA) refers to tumor-derived DNA fragments found in blood and is commonly known as a liquid biopsy. ctDNA has gained interest as a biomarker in early and advanced solid tumors to prognosticate and guide treatment choices (1, 2). The choice of ctDNA assay can vary based on cancer type and intent of use. Real-world use and performance of ctDNA testing relies on clinical validity and utility (3). Herein we review current evidence for the use of ctDNA in localized and advanced solid tumors and future areas of research required for clinical utility.

The potential clinical utility of ctDNA has been explored in both localized and metastatic settings. In early-stage disease, ctDNA has been investigated for early cancer detection, minimal residual disease (MRD) detection, and recurrence surveillance (4). In advanced disease, ctDNA has shown promise in predicting prognosis, monitoring response to treatment, guiding targeted therapeutic approaches, and detecting new resistance mutations. The short half-life of ctDNA enables real-time response assessments that may provide earlier and more accurate results than current radiological assessments (5).

There are several approaches to measuring ctDNA, including next-generation sequencing (NGS), droplet digital polymerase chain reaction, and panel-based assays (6, 7). The methods behind these approaches are beyond the scope of this review. Clinical utility depends on the accuracy of ctDNA analysis, and questions remain around the optimal sampling time and ctDNA quantification method. Biological factors, such as tumor type, stage, disease burden, trauma, and inflammation, can affect results independent of technical assay factors. False negatives can occur in low-shedding tumors or certain anatomic sites such as the peritoneum or central nervous system (CNS), and false positives can occur with clonal hematopoiesis (8).

This review explores the current evidence for the use of ctDNA across all solid tumors in the early and advanced settings for prognostic and predictive purposes in clinical practice (**Figure 1**). The role of ctDNA in screening is not covered but is an important area of ongoing research (9).

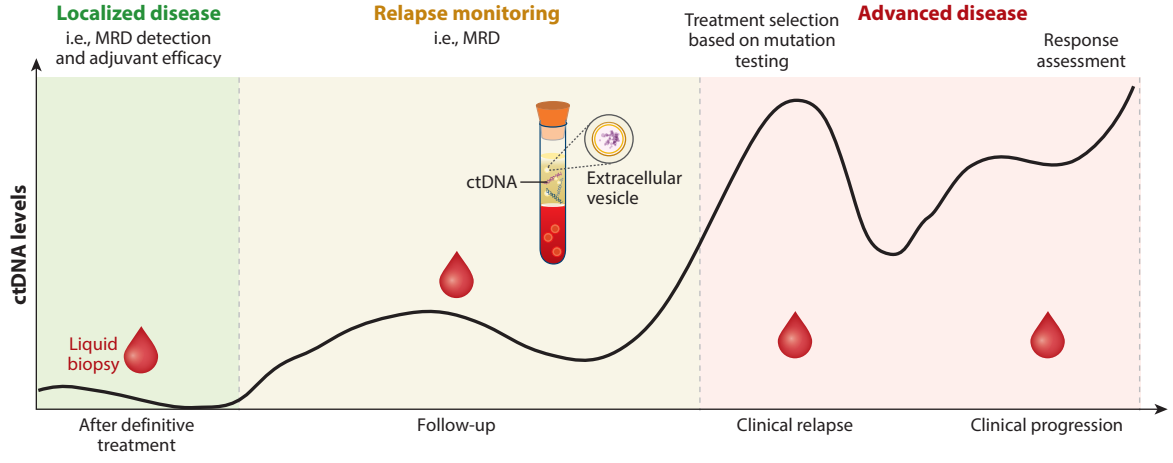


Figure 1

Use of ctDNA in clinical practice. Abbreviations: ctDNA, circulating tumor DNA; MRD, minimal residual disease.

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LOCALIZED DISEASE

Minimal Residual Disease and Prognostication

Surgery is often the definitive treatment for localized solid malignancies, but many patients will experience recurrence (10–12). This is attributed to micrometastases that are not detectable by conventional imaging techniques. It is in this setting that ctDNA has shown potential clinical utility, namely by detecting MRD, which can help stratify patients' risk of relapse to inform the need for further adjuvant treatment (13).

For ctDNA to be incorporated into routine practice, consistently high assay sensitivity and specificity to detect relapse is required (14). High sensitivity is particularly important for a de-escalation treatment approach to minimize undertreatment, while high specificity is critical for treatment escalation to avoid too many false positives and overtreatment. Broadly speaking, there are two approaches to MRD detection using ctDNA, tumor informed and tumor naïve. Tumor-informed assays are based on alterations found in tumor tissue, making them more personalized and providing higher sensitivity, while tumor-naïve assays use a fixed panel to analyze ctDNA in the plasma (15, 16). Tumor-naïve approaches, which do not require prior tumor sequencing, are less costly and have a faster result turnaround, but sometimes these advantages come at the cost of lower sensitivity (15).

Many studies have reported the clinical validity of ctDNA in prognostication for localized solid malignancies and its potential role in treatment algorithms (7). With the exception of the GALAXY study, which included 1,039 patients with resected colorectal cancer (CRC) (17), most of these studies were small observational or retrospective studies. A number of meta-analyses have also been conducted (**Table 1**). A large tumor-agnostic meta-analysis reported that ctDNA exhibits high prognostic accuracy for relapse (82.2% sensitivity, 94.1% specificity) (18). This meta-analysis also reported that sensitivity was much lower (58.3%) when testing with single time points rather than multiple (18). In addition, the prognostic accuracy differed by tumor subgroup (highest in CRC and lowest in lung cancer) (18). All of the studies included in this analysis were observational and varied in design, ctDNA testing methods, and definitions for positivity and MRD.

Despite these limitations, many studies across a spectrum of tumor types have consistently confirmed a strong relationship between tumor-informed ctDNA detection and poor recurrence-free survival (RFS) and overall survival (OS). Garcia-Murillas et al. (19) reported that ctDNA-negative patients with early breast cancer had an 83% decrease in risk of relapse. The GALAXY study investigated ctDNA analysis for detecting recurrence in patients with localized CRC postsurgery. The study reported that patients with ctDNA detected at 4 weeks after surgery (i.e., ctDNA-positive patients) were 12 times more likely to experience recurrence compared to those with undetectable ctDNA (17). Furthermore, it was reported that ctDNA-positive patients experienced a greater benefit from adjuvant chemotherapy compared to ctDNA-negative patients; ctDNA-positive patients who did not receive adjuvant chemotherapy had a significant risk of recurrence (HR, 6.59; 95% CI, 3.53–12.3; $p < 0.001$) (17). ctDNA detection has also been shown to be strongly predictive of recurrence in lung and bladder cancers (20, 21).

Incorporating ctDNA Minimal Residual Disease in the Treatment Paradigm

In current practice, clinicopathological factors are often used to assess the risk of recurrence (22–24). More intense neoadjuvant or adjuvant treatment is typically recommended for patients with adverse features. However, these clinicopathological factors can be imprecise in predicting

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Table 1 Meta-analyses investigating the clinical validity of ctDNA in localized solid malignancies[a]

Authors	Publication year	Sample size	Setting	Sensitivity	Specificity	RFS/DFS	Overall survival
Lung cancer							
Guo et al. (90)	2023	9 studies, 784 patients	MRD, posttreatment surveillance	0.58 (95% CI, 0.49–0.66)	0.93 (95% CI, 0.89–0.96)	HR, 8.32 (95% CI, 4.85–14.28)	HR, 4.73 (95% CI, 2.57–8.70)
Wang et al. (42)	2022	7 studies, 1,138 patients	Posttreatment surveillance	NR	NR	HR 1 week before surgery, 3.63 (95% CI, 2.56–5.14) HR 1–3 months after surgery, 6.28 (95% CI, 2.76–14.32)	HR 1 week before surgery, 3.88 (95% CI, 2.45–5.6.14) HR 1–3 months after surgery, 5.63 (95% CI, 1.44–21.90)
Zhong et al. (91)	2023	11 studies, 1,251 patients	Posttreatment surveillance	0.76 (95% CI, 0.70–0.82)	0.87 (95% CI, 0.84–0.90)	NR	NR
Chen et al. (92)	2023	Baseline: 13 studies, 1,113 patients Post-op: 14 studies, 1,051 patients	Posttreatment surveillance	NR	NR	Baseline HR, 3.06 (95% CI, 2.13–4.41) Post-op HR, 4.74 (95% CI, 3.21–7.02)	NR
Breast cancer							
Papakonstantinou et al. (93)	2022	11 studies, 651 patients	Posttreatment surveillance	0.34 for pathological complete response	0.66 for pathological complete response	HR, 5.67 (95% CI, 2.73–11.75)	HR, 4.00 (95% CI, 1.90–8.42)
Cullinane et al. (94)	2020	8 studies, 739 patients	Posttreatment surveillance	NR	NR	HR, 8.32 (95% CI, 3.01–22.99)	NR
Colorectal cancer							
Chidharla et al. (95)	2023	22 studies, 3,568 patients	Posttreatment surveillance	NR	NR	HR, 7.27 (95% CI, 5.49–9.62)	NR
Wang et al. (96)	2020	7 studies, 424 patients	Posttreatment surveillance	NR	NR	Predicting recurrence: RR, 4.65 (95% CI, 2.68–8.08) HR, 9.14 (95% CI, 4.02–20.75)	NR
Yekedüz et al. (97)	2020	4 studies, 489 patients	Posttreatment surveillance	NR	NR	HR, 5.88 (95% CI, 3.62–9.56)	NR
Chen et al. (98)	2022	7 studies, 821 patients	Posttreatment surveillance	NR	NR	HR, 5.41 (95% CI, 2.37–8.45)	NR
Mi et al. (99)	2022	11 studies[b]	Posttreatment surveillance	NR	NR	HR, 7.26 (95% CI, 5.48–9.62)	NR
Fan et al. (100)	2023	6 studies, 1,713 patients	Posttreatment surveillance	NR	NR	HR, 4.64 (95% CI, 2.17–9.92)	NR
Dawood et al. (101)	2023	14 studies[b]	Posttreatment surveillance	0.68 (95% CI, 0.57–0.77)	0.95 (95% CI, 0.91–0.97)	NR	NR
Mi et al. (102)	2022	7 studies[b]	Posttreatment surveillance	NR	NR	HR, 7.08 (95% CI, 4.12–12.14)	HR, 3.03 (95% CI, 1.86–4.95)

(*Continued*)

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Table 1 (*Continued*)

Authors	Publication year	Sample size	Setting	Sensitivity	Specificity	RFS/DFS	Overall survival
Pancreatic cancer							
Lee et al. (103)	2019	5 studies, 375 patients	Posttreatment surveillance	NR	NR	Baseline HR, 1.96 (95% CI, 0.65–5.87) Post-op HR, 2.20 (95% CI, 0.99–4.87) Overall HR, 1.99 (95% CI, 1.10–3.60)	Baseline HR, 2.27 (95% CI, 1.13–4.56) Post-op HR, 3.66 (95% CI, 1.45–9.28) Overall HR, 2.80 (95% CI, 1.81–4.32)
Fang et al. (104)	2020	12 studies[b]	Pre-op and post-op	NR	NR	NR	HR, 2.95 (95% CI, 2.11–4.12)
Esophagogastric cancer							
Mi et al. (102)	2023	8 studies, 423 patients	Surveillance	NR	NR	HR, 6.37 (95% CI, 2.70–15.01)	HR, 4.58 (95% CI, 1.68–12.49)
Chidambaram & Markar (105)	2022	4 studies, 233 patients	Surveillance	49%	96%	NR	NR
Min et al. (106)	2023	34 studies, 5,091 patients	Surveillance	NR	NR	HR, 3.13 (95% CI, 2.08–4.72)	HR, 2.74 (95% CI, 2.24–3.35)
Other cancers							
Mittal et al. (18)	2023	24 studies, 1,516 patients	Surveillance	82.2%	94.1%	OR, 31.04 (95% CI, 23.95–40.23)	NR
Gögenur et al. (107)	2022	10 studies, 727 patients	Surveillance	NR	NR	RR of recurrence, 8.03 (95% CI, 3.15–20.43)	NR

[a]All meta-analyses included a mix of ctDNA methods. Time point and ctDNA sampling frequency varied across all meta-analyses.

[b]Patient number was not reported.

Abbreviations: CI, confidence interval; ctDNA, circulating tumor DNA; DFS, disease-free survival; HR, hazard ratio; MRD, minimal residual disease; NR, not researched; OR, odds ratio; RFS, recurrence-free survival; RR, risk ratio.

relapse, and thus lack accuracy in identifying patients who would most benefit from more intense treatment, resulting in many patients being overtreated. In addition, some patients who lack these risk factors still relapse. Consequently, there remains a great need to identify more accurate biomarkers.

To this end, randomized trials are required to demonstrate that using ctDNA testing to guide management will lead to improved patient outcomes (e.g., increasing survival, reducing treatment and associated toxicity). The DYNAMIC study was the first prospective multicenter randomized trial to demonstrate the clinical utility of a ctDNA-guided adjuvant treatment approach in resected stage II colon cancer (25). This study randomized 455 stage II colon cancer patients to receive adjuvant therapy based on standard histopathological factors or postoperative ctDNA results. In the ctDNA-guided arm, patients who had a positive ctDNA result using a tumor-informed assay at week 4 or 7 postsurgery received adjuvant chemotherapy, while ctDNA-negative patients underwent surveillance. The study reported that an approach informed by ctDNA results was noninferior to the current standard of care in terms of 2-year RFS (93.5% versus 92.4%). Importantly, the ctDNA approach substantially reduced the proportion of patients exposed to chemotherapy and its associated toxicity (15% versus 28%) (25). The more recent observational BESPOKE trial included 145 stage II and 196 stage III colon cancer patients and demonstrated the benefit of adjuvant therapy in MRD-positive patients but not in MRD-negative patients (HR, 1.1; 95% CI, 0.3–3.9; $p = 0.89$) (26).

The COBRA study (NCT04068103) is investigating ctDNA detection (i.e., tumor-naïve assays) for treatment escalation by randomizing patients with clinically and histopathologically low-risk stage IIA colon cancer who are deemed not suitable for adjuvant chemotherapy to observation or ctDNA testing, with ctDNA-positive patients receiving adjuvant chemotherapy (27). The aim is to demonstrate that ctDNA-positive patients will have a higher ctDNA clearance rate and better RFS if treated with adjuvant chemotherapy compared to surveillance alone. Unfortunately, this study ceased recruitment prematurely after an interim futility analysis of the first 16 ctDNA-positive patients showed an unexpected higher ctDNA clearance rate with surveillance than with chemotherapy (28). Assay specificity has been speculated as the reason for these unusual results, but critical information regarding recurrence outcomes, which will help confirm this observation, is not yet available.

ctDNA for Assessing Neoadjuvant and Adjuvant Treatment Efficacy

There is evidence that ctDNA clearance or persistence can be a surrogate for treatment efficacy in localized settings across multiple solid cancers. Zhou et al. (29) reported that breast cancer patients with persistent mid-treatment ctDNA were significantly more likely to have a noncomplete pathological response. This has similarly been reported for bladder cancer, with ctDNA clearance being correlated with complete pathological response (30). In addition, non-small cell lung cancer (NSCLC) patients with persistent ctDNA detection have been reported to have worse clinical outcomes (31). Consequently, a number of studies have focused on developing strategies and escalating treatments to improve outcomes for these patients.

The PEGASUS trial evaluated ctDNA-guided treatment after surgery compared to the standard approaches for colon cancer patients. The trial showed that a chemotherapy escalation strategy based on ctDNA positivity allowed 48% of patients to remain relapse free through a potential second-line salvage adjuvant approach (32). Similarly, the ALTAIR study, which is investigating trifluridine and tipiracil for patients with ctDNA detected within 2 years after surgery and adjuvant chemotherapy, aims to evaluate treatment escalation strategies (33). The results are expected to be reported in the near future.

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ctDNA for Relapse Monitoring and Surveillance

In current practice, conventional imaging through computed tomography or positron emission tomography is the mainstay modality for surveillance. The advantage of conventional imaging is that it provides precise measurements and locations of lesions. However, lesions need to be large enough to be detected, and often when they are detected patients may no longer be curable due to considerable lag time. To help improve the sensitivity for detecting lesions, tumor markers, such as carcinoembryonic antigen (CEA), cancer antigen 125 (CA125), CA15.3, and CA19.9, are also often incorporated into surveillance schedules for patients. However, these tumor markers are insensitive and have high false-positive and -negative rates. In fact, when compared to ctDNA, they consistently underperform. CEA sensitivity ranges between 41% and 67% for colon cancer (34), CA125 sensitivity ranges between 30% and 60% for ovarian cancer (35), and CA19.9 sensitivity ranges between 60% and 80% for pancreatic cancer (36). On the contrary, ctDNA has repeatedly shown improved lead times, higher sensitivity, and higher specificity, making it an overall more powerful tool compared to tumor markers (34, 37–39). The ctDNA strategy as a posttreatment surveillance tool compared to imaging is being investigated through a randomized controlled trial, IMPROVE-IT2 (40). In fact, the lead time for ctDNA detection of recurrence ranges between 3 and 11 months prior to clinical or radiological recurrence, highlighting its potential (18, 41, 42).

While results from other prospective studies of other tumor groups (outside of colon cancer) are lacking, a similar approach could be incorporated for nearly all localized solid tumors. Several prospective studies are investigating this approach in breast cancer (NCT04849364, NCT04434040), bladder cancer (NCT04138628, NCT04660344), ovarian cancer (NCT05212779), lung cancer (NCT04385368, NCT04585477, NCT05536505, NCT04642469, NCT05460195), and melanoma (NCT04901988).

ADVANCED DISEASE

ctDNA Mutation Detection for Targeted Therapy

Recommendations for targeted therapy in advanced solid tumors often rely on NGS panels, both tumor specific and tumor agnostic (43). Liquid biopsy, specifically ctDNA mutation detection, is increasingly utilized when obtaining tumor tissue proves challenging. International guidelines endorse the use of liquid biopsy to identify several actionable mutations, yet its clinical utility hinges on reliability, and using tissue remains the favored approach if available (44).

Several cancers have demonstrated the effectiveness of using ctDNA to detect mutations for targeted therapy, with US Food and Drug Administration (FDA) approval supporting several of these applications. For epidermal growth factor receptor (EGFR)–mutant NSCLC, the cobas EGFR test received FDA approval based on evidence that ctDNA demonstrates high concordance and sensitivity (70%) for detecting relevant mutations, including *T790*, one of the resistance path mechanisms (45, 46). Alterations in *BRCA* have been targeted by poly (ADP-ribose) polymerase (PARP) inhibitors across breast, gynecological, pancreatic, and prostate carcinomas; however, it is only in prostate carcinoma where high concordance between plasma and tissue testing supports the FDA approval of FoundationOne liquid testing (47).

Many trials have shown promise for the use of ctDNA in guiding therapy. In breast cancer, the BOLERO-2 trial demonstrated 81.6% concordance for *PIK3CA* mutations between tissue and plasma (48). For *ESR1* mutations that can guide hormonal therapy in advanced breast cancer, ctDNA outperforms tissue testing in analytical sensitivity (49). In pancreatic adenocarcinoma, a cancer where tissue can be particularly challenging to obtain, a trial demonstrated 73% concordance for *KRAS* mutations between tissue and ctDNA, which is especially relevant with the development of potential new therapies targeting *KRASG12C* and *KRASG12D* (50). The

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GOZILA trial, which studied patients with gastrointestinal cancer, showed that ctDNA had a higher success rate for variant detection than tumor tissue and shortened the time to receive results (median 7 versus 19 days), improving the trial enrollment rate (9.5% versus 4.1%; $p < 0.0001$) (51).

ctDNA may have less potential utility in some tumor types where detection rates are low. Renal cell carcinoma (RCC) is considered a low-shedding cancer, and the von Hippel-Lindau driver is detected in 71.8% of tumor tissue samples but only 25% of ctDNA samples (52). Furthermore, when using the MSK-IMPACT platform in 110 patients with metastatic clear cell RCC, concordance between tissue and ctDNA variants was notably low (53). This challenges the reliability of ctDNA testing in RCC and underscores the need for alternative approaches for therapeutic target identification. The limitations of ctDNA in CNS tumors are also notable. When analyzing *IDH1*, a recognized druggable target in this context, low concordance between tissue and ctDNA has been observed (54). Consequently, the current recommendation is for tissue-based testing for CNS tumors. It is also worth noting that certain metastatic disease sites, such as the lung and peritoneum, are more likely to yield noninformative ctDNA results due to a low ctDNA tumor fraction, and tissue testing would be preferable for patients with metastases only in these sites.

ctDNA in Prognostication and Response Assessment

Prognostication is pivotal in oncological management. Knowledge of prognostic factors can help determine treatment intent, whether curative or noncurative, and guide the choice of systemic therapy, ranging from single-agent to doublet or triplet combinations. The presence of ctDNA in patients' blood is linked to a poorer prognosis across multiple advanced cancer types (**Table 2**). Most studies reported shorter OS and progression-free survival (PFS) with ctDNA positivity (55).

An early indicator or marker of therapy response is crucial for maximizing benefit and minimizing toxicity. ctDNA is emerging as a real-time biomarker for treatment response with the potential to complement radiological response assessment (5, 56–58). A systematic review of 10 studies using immunotherapy in the advanced NSCLC setting showed that early reduction in ctDNA was associated with improvements in OS (HR, 0.10; 95% CI, 0.10–0.35), PFS (HR, 0.30; 95% CI, 0.22–0.41), and overall response rate (ORR) (odds ratio, 0.07; 95% CI, 0.03–0.18) (59). The large phase III IMpower150 trial also demonstrated that changes in ctDNA at week six were associated with radiographic response and OS, making ctDNA a potential early predictive biomarker (60). A meta-analysis of 2,823 patients with CRC revealed a strong correlation between ctDNA detection after treatment and PFS (HR, 3.15; 95% CI, 2.10–4.73; $p < 0.00001$) (61). Changes in ctDNA after one cycle of chemotherapy were also correlated with response, demonstrating a time advantage over radiological assessment (61). While there is increasing evidence to show that ctDNA decline correlates with improved response and survival, there are fewer published studies demonstrating ctDNA use to identify patients who are not responding. In CRC, the meta-analysis by Callesen et al. (62) showed that a smaller or no decrease in early ctDNA was a marker of reduced response and a negative prognosis; however, there is a need for interventional studies using ctDNA to initiate accelerated decision-making and therapy escalation or therapy switch.

For targeted therapies, studies have shown the potential of ctDNA clearance as an early predictive biomarker of response. In NSCLC with known targetable mutations, an early reduction of ctDNA in patients treated with tyrosine kinase inhibitors correlated with improved PFS (pooled HR, 2.71; 95% CI, 1.85–3.65) (63). In EGFR-directed therapy for NSCLC, ctDNA clearance after 3 weeks correlated with longer PFS compared to no ctDNA clearance (median 13.5 versus 9.5 months) (58). In *ALK* fusion-positive patients, targeted therapy led to decreased ctDNA (using a threshold of ≤ 0) at week four, which correlated with complete or partial response, suggesting

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Table 2 Meta-analyses investigating the utility of ctDNA as a prognostic marker in advanced solid malignancies

Authors	Publication year	Sample size	Cancer	ctDNA time point and metrics[a]	OS	Outcome
Sun et al. (55)	2023	22 studies, 723 patients (OS) and 22 studies, 622 patients (PFS)	Non-small cell lung cancer	Variable Detectable versus undetectable	HR, 2.55 (95% CI, 1.89–3.45)	PFS HR, 2.34 (95% CI, 1.89–2.89)
Zaman et al. (63)	2023	11 studies, 1,359 patients	Non-small cell lung cancer	Detectable versus undetectable at baseline/pretreatment	HR, 1.15 (95% CI, 0.85–1.45)	PFS HR, 1.35 (95% CI, 0.83–1.87)
Wang et al. (59)	2021	3 studies, 625 patients	Non-small cell lung cancer	Detectable versus undetectable at baseline	HR, 1.18 (95% CI, 0.93–1.51)	PFS HR, 0.98 (95% CI, 0.8–1.21)
Jones et al. (61)	2021	28 studies, 2,823 patients	Colorectal cancer	Pre- and posttreatment (exacts variable) Detectable versus undetectable	HR, 2.2 (95% CI, 1.79–2.69)	PFS HR, 3.15 (95% CI, 2.10–4.73)
Callesen et al. (62)	2022	2 studies, 161 patients (OS) and 10 studies, 569 patients (RFS)	Colorectal cancer (receiving locoregional therapy)	Postablation detection/threshold variable	HR, 7.5 (95% CI, 2.0–27.3)	RFS HR, 4.5 (95% CI, 3.4–6.1)
Milin-Lazovic et al. (108)	2021	15 studies, 1,072 patients (OS) and 7 studies, 378 patients (PFS)[b]	Pancreatic cancer	Pretreatment *KRAS* detection positive versus negative	HR, 2.51 (95% CI, 1.90–3.31)	PFS HR, 2.51 (95% CI, 1.98–3.19)
Gao et al. (109)	2017	10 studies, 521 patients	Gastric cancer	ctDNA presence at baseline	HR, 1.77 (95% CI, 1.38–2.28)	DFS HR, 4.69 (95% CI, 2.23–9.85)
Cullinane et al. (94)	2020	3 studies, 261 patients	Breast cancer	ctDNA level pre- and posttreatment	NR	RFS HR, 1.91 (95% CI, 1.35–2.71)
Gracie et al. (80)	2021	19 studies, 1,349 patients	Melanoma	Baseline ctDNA detection positive versus negative	ddPCR: HR, 2.78 (95% CI, 2.21–3.49) Non-ddPCR: HR, 2.58 (95% CI, 1.74–3.84)	ddPCR: PFS HR, 2.1 (95% CI, 1.71–2.59) Non-ddPCR: PFS HR, 2.15 (95% CI, 1.53–3.41)
Feng et al. (110)	2021	9 studies, 617 patients	Melanoma	Detectable versus undetectable baseline ctDNA	HR, 2.91 (95% CI, 2.22–3.82)	PFS HR, 2.75 (95% CI, 1.98–3.83)
Mehra et al. (111)	2018	2 studies, 571 patients	Prostate cancer	Baseline cfDNA concentration quartiles	HR, 1.55 (95% CI, 1.15–2.09)	Radiological PFS HR, 1.56 (95% CI, 1.08–2.24)

[a]All meta-analyses included a mix of ctDNA methods.

[b]These studies included locally advanced and metastatic patients.

Abbreviations: cfDNA, cell-free DNA; CI, confidence interval; ctDNA, circulating tumor DNA; ddPCR, droplet digital polymerase chain reaction; DFS, disease-free survival; HR, hazard ratio; NR, not researched; OS, overall survival; PFS, progression-free survival; RFS, recurrence-free survival.

that ctDNA may provide an earlier assessment of response compared to conventional radiological assessment (64). Similarly, the presence of *BRAF V600E* clearance (>90% decrease on droplet digital polymerase chain reaction) in CRC during targeted therapy at 6 weeks also correlated with radiology response (65, 66). The >90% reduction in *BRAF V600E* ctDNA has been shown to correlate with response, while CEA has not, suggesting that serial ctDNA assessment of specific molecular targets may be a future biomarker to predict earlier response or resistance in CRC (67). Human epidermal growth factor receptor 2 (HER2) is the most common targetable option in

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gastric cancer and has been used as a marker of response using ctDNA. Patients with advanced HER2-positive gastric cancer with ctDNA clearance at 9 weeks of treatment with pembrolizumab and trastuzumab had improved PFS compared to those without ctDNA clearance (12.3 versus 3.9 months; $p = 0.02$) (68). In a cohort of 35 patients with gastric cancer treated in the first-line setting, those with a >50% decline in maximum variant allele frequency (VAF) via NGS had a superior OS compared to those with a <50% decline in maximum VAF (13.7 versus 8.6 months; $p = 0.02$; HR, 0.3; 95% CI, 0.1–0.8) (69).

Prostate cancer response is often monitored by serial prostate-specific antigen (PSA) measurements. Radiological response assessments are also used but can be challenging with predominant nodal or skeletal disease. A study of 43 patients with metastatic castrate-resistant disease found that an increase in ctDNA was strongly associated with radiological and PSA progression but not with isolated PSA increases or radiological flares; this suggests that ctDNA could help prevent premature treatment cessation or changes based on PSA or radiological flare phenomena, allowing for more accurate adjustments in cases of true disease progression (70). The extent of ctDNA decline may also be relevant for prognostication, but exact cutoffs have not been defined (71, 72). In genitourinary cancers, a longitudinal study across 15 renal, 4 urothelial, and 1 prostate cancer patient showed 83% concordance between radiological and ctDNA response and immune checkpoint inhibitors, while discordance occurred in three patients (all of whom had RCC) (73). Among the discordant patients, two developed CNS disease. Overall, these data suggest that larger-scale studies are needed to determine if ctDNA is a reliable biomarker in RCC (73–75).

In advanced breast cancer, trials like PALOMA-3 and BEECH demonstrated improved PFS with reduced *PIK3CA* on liquid biopsy (76, 77). Gynecological cancer studies of ctDNA for prognostication are limited but have shown promise in high-grade serous ovarian carcinoma, with >60% reduction in *TP53* VAF during chemotherapy being associated with improved PFS (78). A study of 420 patients with CRC, ovarian cancer, or NSCLC retrospectively reviewed ctDNA levels before treatment and at the first evaluation of response; the findings showed a moderate correlation between ctDNA response and ORR ($R^2 = 0.57$), with ctDNA demonstrating better performance than radiographic ORR for predicting OS (79).

Immunotherapy is the international standard of care for metastatic melanoma. Pseudoprogression is a key limitation of radiological assessments of response, as infiltration of immune cells can lead to the enlargement or presence of new lesions on radiological imaging. This has been reported in 2–10% of melanoma patients, and repeat imaging no less than 4 weeks later is used to determine if the changes are true or pseudoprogression (80, 81). Examination of longitudinal ctDNA has been used to differentiate radiological true progression from pseudoprogression, with ctDNA demonstrating a sensitivity of 90% (95% CI, 68–99%) and specificity of 100% (95% CI, 60–100%) in predicting pseudoprogression (82). In melanoma and lung cancer, CNS progression can be problematic. ctDNA detection for primary and metastatic CNS disease is challenging, and studies suggest that use of cerebrospinal fluid may enhance sensitivity but is less practical for longitudinal assessments of response (83, 84).

Emerging roles of ctDNA include identifying the emergence of resistance mutations and managing early resistance (85, 86). In NSCLC, emergence of *T790* and the mesenchymal epithelial transition factor receptor (*MET*) gene can be detected with a high rate of match to tissue to predict resistance via a method that is less invasive than traditional tissue biopsy (87). The CHRONOS trial used ctDNA to detect EGFR inhibitor-resistance (i.e., *RAS*, *BRAF*, and *EGFR* extracellular domain mutations) and enrolled patients without these mutations for anti-EGFR rechallenge therapy. By enriching patients without resistant mutations, anti-EGFR rechallenge resulted in a response rate of 30%, suggesting that ctDNA could successfully be used to guide

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therapy rechallenge (88). Similarly, the PADA-1 trial, which focused on advanced breast cancers, showed that ctDNA monitoring of *ESR1* mutations successfully predicted the need for a therapy switch, leading to improved PFS (89).

Overall, ctDNA holds promise in prognostication and response assessment across various cancers, with ongoing research exploring its full potential in clinical practice. Interventional randomized trials in the advanced cancer setting are warranted across multiple tumor types, and several are underway to establish the clinical utility of ctDNA in response assessment, treatment initiation, and modification: Rapid 1 (NCT04786600), TACT-D (NCT03844620), NCT04831528, MONDRIAN (NCT04720729), FAIM (NCT04920708), NCT05770531, NCT05826964, PROTRACT (NCT04015622), PRELUCA (NCT05889247), and FLAME (NCT04769388).

FUTURE PERSPECTIVES

Mounting evidence supports the potential of qualitative and quantitative ctDNA analysis to assist prognostication and predict therapy responses across the entire treatment paradigm. Beyond its use in genotyping to identify patients for targeted therapies and its emerging role in guiding adjuvant therapy in stage II colon cancer, the routine clinical application of ctDNA remains limited due to the lack of randomized trials demonstrating its clinical utility. Additional challenges for the widespread use of ctDNA include cost; tumor biology, with low-shedding tumors or certain disease sites causing false-negative or noninformative ctDNA results; and the lack of effective therapies (83). To advance ctDNA into broad clinical use to guide treatment changes, interventional prospective trials are urgently needed. It is also essential to focus on the prospective standardization and validation of preanalytic and analytic methods and time points for ctDNA analysis, which may vary depending on the aim of testing and tumor type. Given the rapid pace at which ctDNA technology is advancing and the prolonged course of adjuvant trials, biobanking of samples is imperative to allow retrospective testing of the latest assays so that patients can benefit from scientific advances in a timely manner. Clear and standardized definitions and cutoffs are required for ctDNA molecular response and progression in the advanced setting. For the detection of targetable alterations, prospective studies are required to assess concordance between tissue and plasma testing, ensuring the reliability of these tests for making treatment decisions in clinical practice. Based on current evidence, it is unlikely that ctDNA will replace current staging or biomarkers, but it will complement existing clinicopathological features in refining risk stratification algorithms, enabling further precision in cancer medicine in this era of treatment intensification across oncological practice.

CONCLUSION

ctDNA has already proven to be a pan-cancer biomarker for prognostic and predictive purposes. Although multiple studies show promising results for the use of ctDNA to determine MRD and early recurrence in the localized setting and guide therapy decisions, response assessments, and prognostication in the advanced setting, larger prospective interventional studies are needed before we can reliably use ctDNA to guide practice. A one-size-fits-all approach will not suffice, and each assay and intended use for each tumor type will need to be tested individually with a randomized trial prior to clinical adoption.

DISCLOSURE STATEMENT

J.T. has speaking, advisory, or consultancy roles at Haystack Oncology, Amgen, Novartis, AstraZeneca, Merck Serono, Merck Sharp & Dohme, Pierre Fabre, Bristol Myers Squibb, Daiichi Sankyo, Gilead, Illumina, BeiGene, Roche, Takeda, and Arcus Biosciences.

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